StudySync — Peer Matching & Collaborative Study Group Platform

UML Design Models: Sequence, Class, Activity & State Diagrams

Course: Software Engineering & Project Management

Project Group: Group 15 (Section F)

Target System: StudySync Web Platform

Date of Release: October 16, 2026

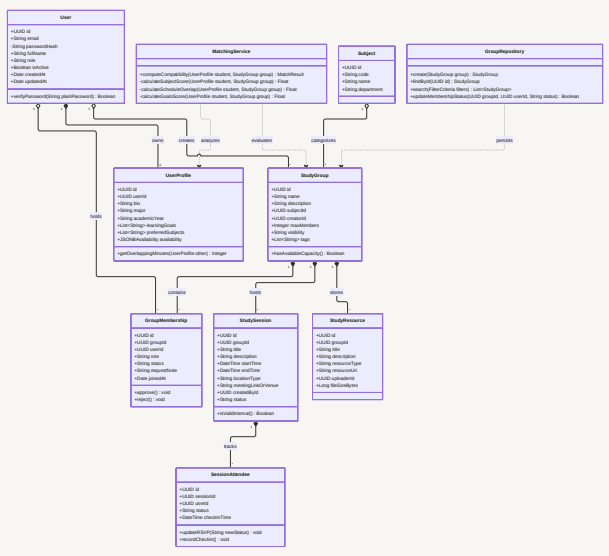
Document Standard: Unified Modeling Language (UML 2.5 Adapted)

1. Overview & UML Modeling Scope

This document provides formal **Unified Modeling Language (UML 2.5)** specifications for the StudySync platform. In accordance with Section 3.C and Section 8 of the course evaluation requirements, this design package documents:

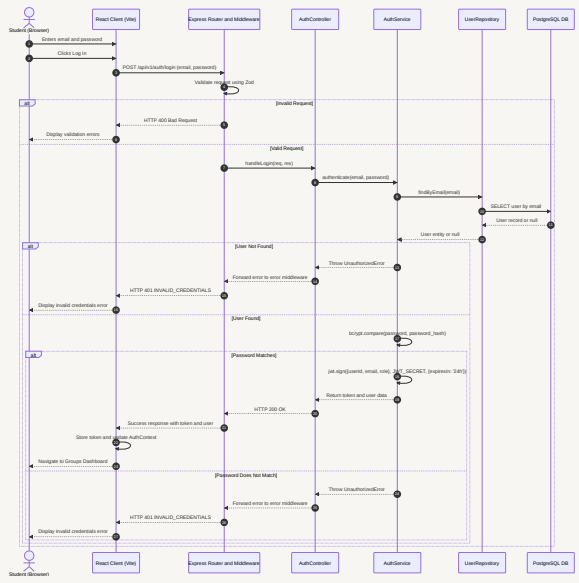
- **Structural Models:** Object-oriented software and domain class diagrams detailing types, method signatures, visibility modifiers, and layer relationships.
- **Dynamic Interaction Models:** Chronological sequence diagrams illustrating actor-to-client-to-server-to-database message flows.
- **Behavioral Process Models:** Activity workflows with validation forks, error branches, and decision boundaries.
- **Lifecycle Models:** Discrete state transitions capturing entity progression through the system.

2. Master Domain & Software Class Diagram

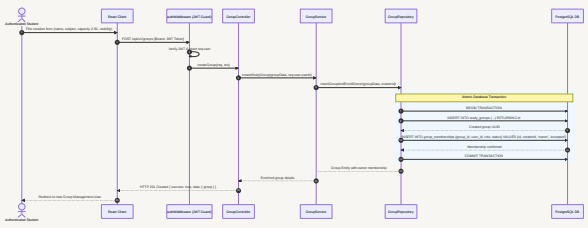


3. UML Sequence Diagrams (Key Interaction Workflows)

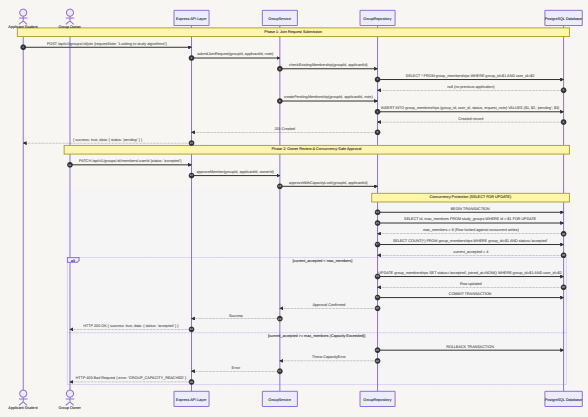
3.1 Sequence 1: Student Authentication & Stateless JWT Issuance



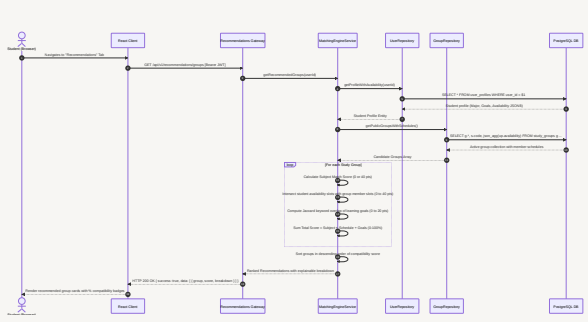
3.2 Sequence 2: Study Group Creation & Automatic Owner Enrollment



3.3 Sequence 3: Join Request Submission & Atomic Owner Approval with Row Locking

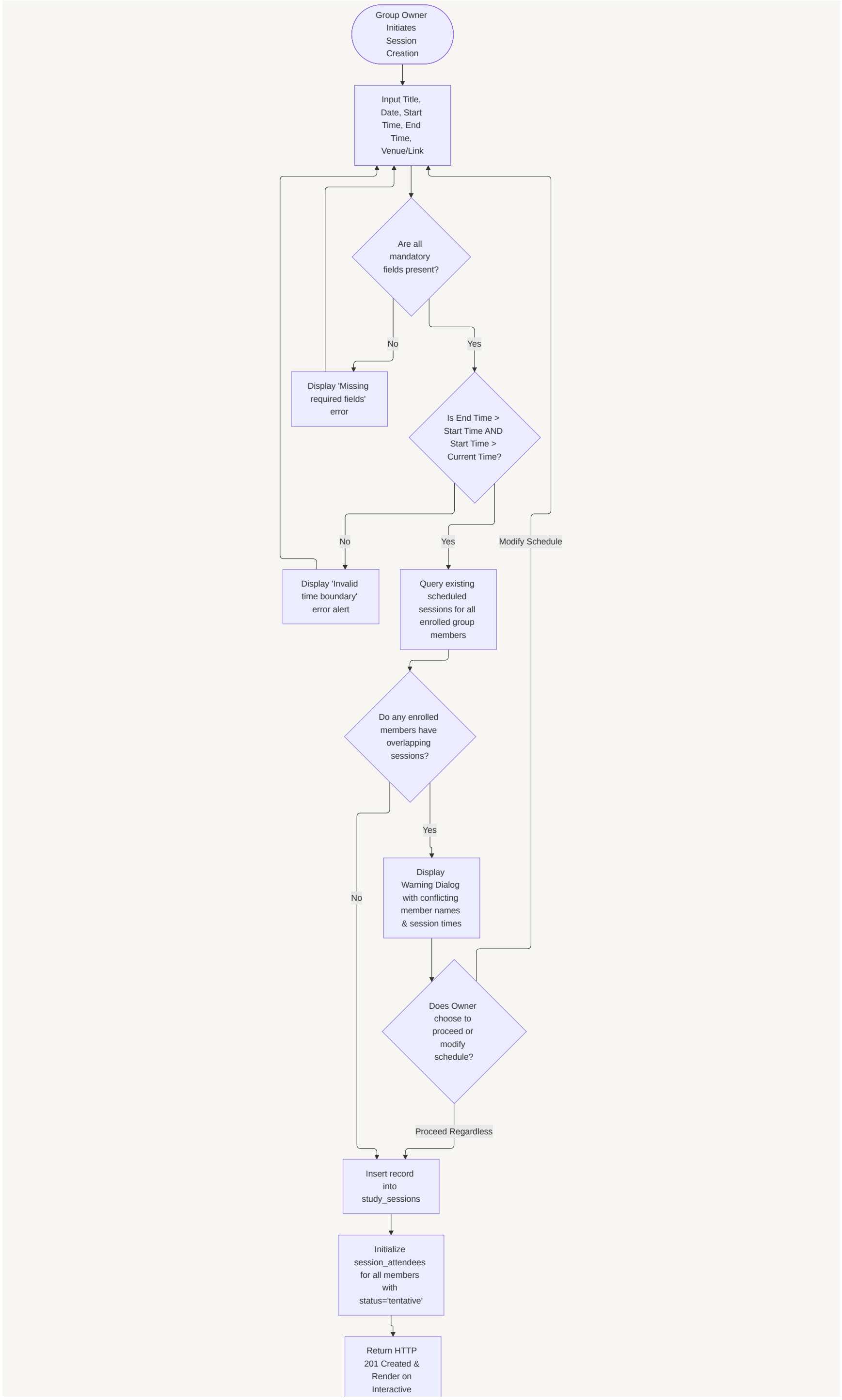


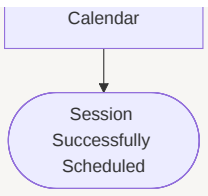
3.4 Sequence 4: 3-Factor Heuristic Matching Engine & Recommendation Flow



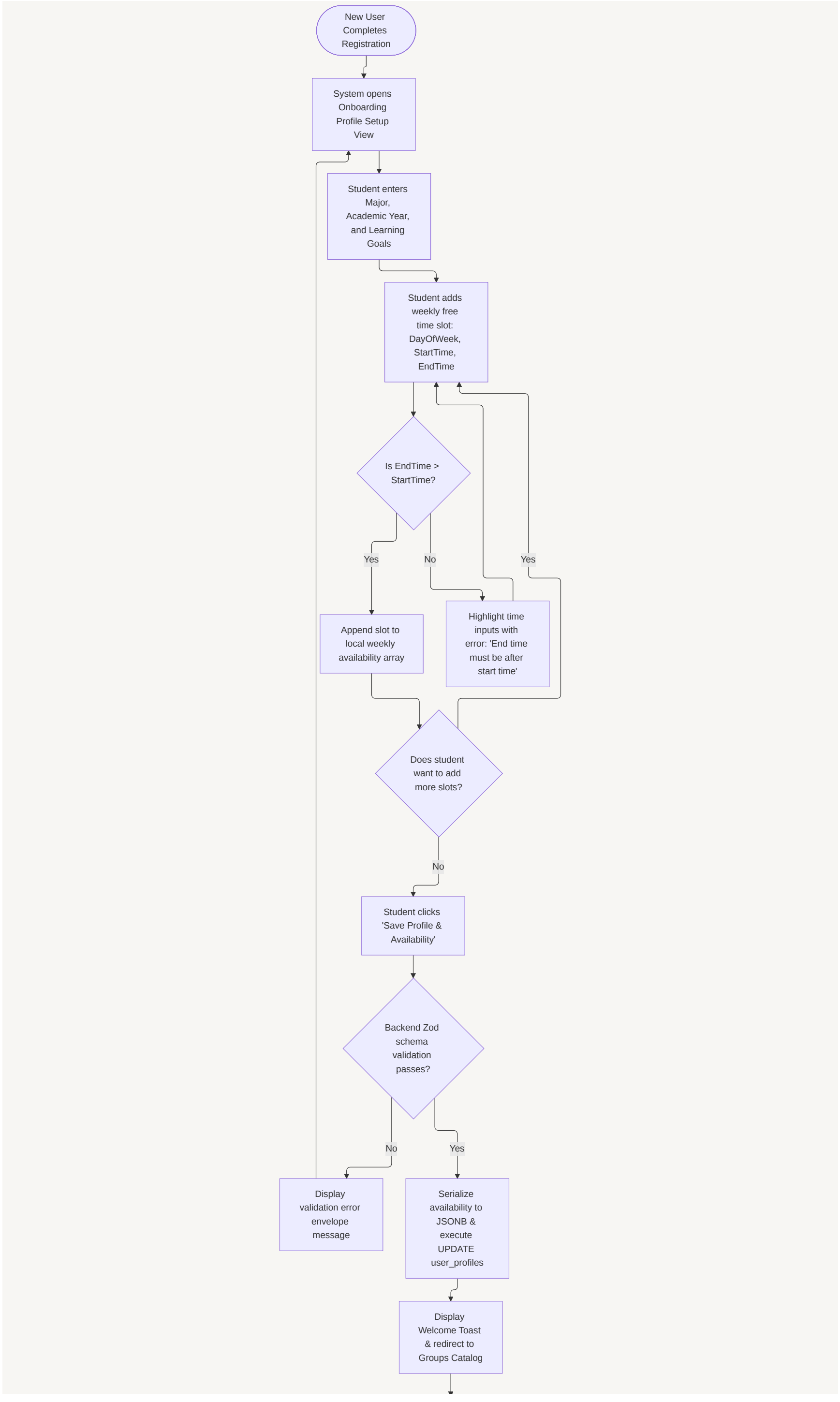
4. UML Activity Diagrams (Operational Business Logic)

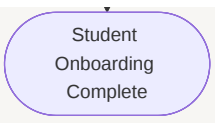
4.1 Activity 1: Study Session Scheduling with Schedule Collision Guard


Group Owner Initiates Session Creation
Input Title, Date, Start Time, End Time, Venue/Link
Are all mandatory fields present?
No
Yes
Display 'Missing required fields' error
Is End Time > Start Time AND Start Time > Current Time?
No
Yes
Modify Schedule
Display 'Invalid time boundary' error alert
Query existing scheduled sessions for all enrolled group members
Do any enrolled members have overlapping sessions?
Yes
No
Display Warning Dialog with conflicting member names & session times
Does Owner choose to proceed or modify schedule?
Proceed Regardless
Insert record into study_sessions
Initialize session_attendees for all members with status='tentative'
Return HTTP 201 Created & Render on Interactive



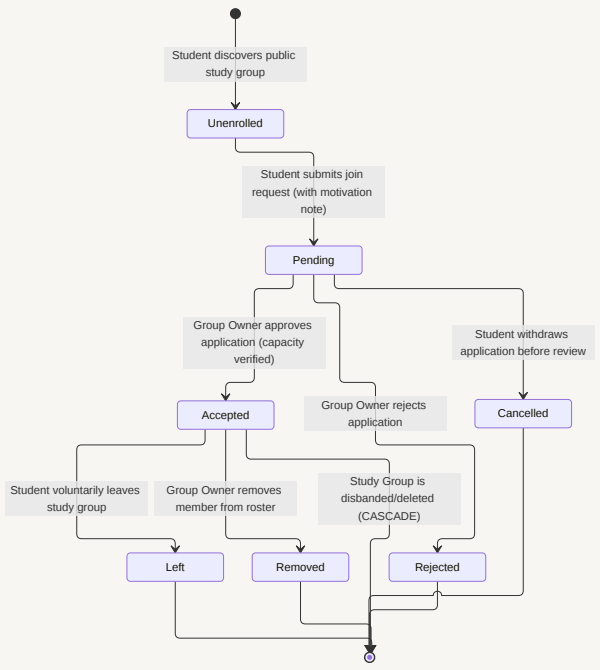
4.2 Activity 2: Student Onboarding & Recurring Weekly Availability Mapping


New User Completes Registration
System opens Onboarding Profile Setup View
Student enters Major, Academic Year, and Learning Goals
Student adds weekly free time slot: DayOfWeek, StartTime, EndTime
Is EndTime > StartTime?
Yes
No
Yes
Append slot to local weekly availability array
Highlight time inputs with error: 'End time must be after start time'
Does student want to add more slots?
No
Student clicks 'Save Profile & Availability'
Backend Zod schema validation passes?
No
Yes
Display validation error envelope message
Serialize availability to JSONB & execute UPDATE user_profiles
Display Welcome Toast & redirect to Groups Catalog



5. UML State Machine Diagrams (Entity Lifecycle States)

5.1 State Machine 1: Group Membership Lifecycle



5.2 State Machine 2: Study Session & Attendance RSVP Lifecycle

